

CLOUD SECURITY PROJECTS

COMPLETE THIS WEEKEND

3 BEGINNER-FRIENDLY
HANDS-ON PROJECTS



Learn
Cloud Security



Understand
Cloud Threats



Hands-on
Monitoring



Real Cloud
Investigations



See What Cloud
Security Engineers
Do Daily



COMPLETE IN
2 DAYS
(WEEKEND PLAN)



BEGINNER TO
INTERMEDIATE
LEVEL



SIMPLE STEPS
EASY TO
UNDERSTAND



LEARN TODAY, **SECURE TOMORROW!**



@blueteamers.io

BEFORE WE START...



Let's understand why Cloud Security is important!

Imagine this...

You have many important things in your life.

Where do they live?

CLOUD
?

You have:



Mobile Phone



Photos



Videos



Documents



Other Files

Sometimes they live inside your phone.

But many times they live inside CLOUD.



That's why Cloud Security exists.



Cloud Security means keeping your data, applications, and systems safe in the cloud from bad guys!



CHOTA SA JOKE 😊



Maa:
Why are you always on that laptop?



You:
I'm securing the cloud!

Maa:
Good. Then secure the water tank also!



By the end of this notebook you'll understand:

- ✓ What cloud security is
- ✓ What cloud security engineers do
- ✓ Common cloud threats
- ✓ Real-world cloud investigations
- ✓ How to protect cloud data



Remember:

Cloud is someone else's computer.
Security is our responsibility.
Learn today, protect tomorrow!



Small steps today,
Big impact in future!



2 OUR WEEKEND MISSION

In just 2 days, we will complete 3 awesome cloud security projects!



Let's turn Saturday and Sunday into a **Cyber Security Adventure!** 😎



≡ SATURDAY ≡

 **MORNING** (9 AM - 1 PM)

PROJECT 1

AWS Cloud Security Monitoring Lab

You will learn:

- What are logs?
- How to monitor cloud activity
- How to investigate user actions
- How CloudTrail works

 **EVENING** (3 PM - 8 PM)

PROJECT 2

Public S3 Bucket Data Leak Lab

You will learn:

- What is misconfiguration?
- How data leaks happen
- How to find public access
- How to fix security issues

≡ SUNDAY ≡

 **MORNING** (9 AM - 1 PM)

PROJECT 3

Cloud Threat Detection Lab

You will learn:

- What are security threats?
- How attackers behave in cloud
- How GuardDuty detects threats
- How to investigate alerts
- How to write incident report

 **AFTERNOON** (3 PM - 6 PM)

MISSION COMPLETE

You will have:

- Completed 3 real-world projects
- Understood cloud security basics
- Gained confidence
- Built your first portfolio projects!



Saturday = Build & Break
(But in a safe way 😊)



By the end of Sunday, you will think like a **Cloud Security Engineer!**

IMPORTANT

Make mistakes.
Ask questions.
Learn a lot.
Enjoy the journey! ❤️



Remember:

Practice today, Protect tomorrow!



Let's Go! 



AWS CLOUD SECURITY MONITORING LAB

Become a Cloud Security Analyst for one day! 😎



MISSION

Your job is to:

- ✓ Watch cloud activity
- ✓ Find suspicious actions
- ✓ Investigate what happened
- ✓ Understand how logs help us



REAL LIFE EXAMPLE

Imagine your school principal wants to know:

- Who entered the classroom?
- Who changed the notice board?
- Who broke the projector?

Cloud logs do the same thing!



WHAT WILL YOU BUILD?

Users / Actions



You or IAM user performs actions in AWS.



CloudTrail



CloudTrail records every action like a CCTV camera.



CloudWatch



CloudWatch helps us see logs, metrics and create alerts.



Alerts



If something suspicious happens, we get an alert!



Investigation



We investigate logs and find out what happened.

Welcome to the world of Cloud Security! 🛡️

WHAT YOU WILL LEARN



- ✓ What is CloudTrail and why it is important
- ✓ How to enable CloudTrail
- ✓ How AWS logs your every action
- ✓ How to find and read logs
- ✓ How to investigate user activities
- ✓ How security teams detect suspicious actions



★ Important:

Logs are your best friends in cloud security. Without logs, you are BLIND! 🙈

TOOLS WE WILL USE



AWS CloudTrail
Records API calls and user activities.



Amazon CloudWatch
View logs, create alarms and dashboards.



S3
Store CloudTrail logs in a secure bucket.



IAM
Create users and give permissions (like real life).

CHOTA SA JOKE 😄

Friend: Bro, what do you do?

You: I check logs all day!

Friend: Boring!

You: Without me, your data will go missing! 😎

HA HA HA!



TIME REQUIRED

4 - 6 Hours

(Take breaks. Drink chai ☕)



GOAL

By the end of this project, you will be able to monitor and investigate cloud activities like a PRO! 💪



Let's put on our **Analyst Hat**



and start the adventure! ➡️

CREATE AWS ACCOUNT

Our first step is to get our own cloud playground!



Think of **AWS** like a big computer shop in the sky. You can rent computers, storage, databases and many other things!

WHAT IS AWS?

AWS (Amazon Web Services) gives us powerful computers, storage and services over the internet.



No need to buy expensive servers



Pay very less (Free Tier available!)



Use from anywhere in the world



Secure and trusted by millions

STEP-BY-STEP: CREATE AWS ACCOUNT

1 Go to

aws.amazon.com

Open your browser and visit the link.

2 Click on

Create an AWS Account

You'll find this button on the top right.

3 Enter your email address



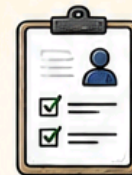
Use your personal email id.

4 Verify your email



AWS will send a code. Enter it to verify.

5 Fill your details



Name, address, phone number etc.

6 Add payment method



Don't worry! You won't be charged if you stay in Free Tier limits.

✓ After successful signup, you will get access to the **AWS Management Console**.

WHAT IS AWS FREE TIER?

AWS gives free services for 12 months to new users. It's like a **FREE TRIAL PERIOD!**

Some popular free services we can use:



EC2 (Virtual Server) – 750 hours/month



S3 (Storage) – 5 GB Standard



RDS (Database) – 750 hours/month



And many more...

INDIAN STYLE JOKE

You: Mom, I need servers for my project.

Mom: Kitne ke aate hain?

You: 1 lakh ka aayega!

Mom: Ohhh... fir kya karega?

You: Mom, **AWS Free Tier** hai na!

Mom: Wah beta! Ab bana apna cloud ka mahal!

Budget friendly beta!



IMPORTANT TIPS

- Always use IAM users, don't use Root user daily.
- Keep your password and Access Keys safe.
- Enable **MFA** (Multi-Factor Authentication).
- Logout when not in use.



MISSION CHECKPOINT

Awesome! You have created your own AWS account. Now we are ready to explore and secure the cloud!



Aaj ka kaam ho gaya! Chai pi lo aur next step ke liye ready ho jao!



ENABLE CLOUDTRAIL

Let's turn ON the CCTV camera of AWS!



CloudTrail is like a CCTV camera that records every single activity in your AWS account.

WHAT IS CLOUDTRAIL?

AWS CloudTrail is a service that records API calls and actions made in your AWS account. It helps us to:



See who did what



See when it happened



See from where



Use logs for security analysis

Simple words: CloudTrail = Your Cloud Diary



WHY IS IT IMPORTANT?

- Tracks user logins
- Tracks changes in resources
- Helps in investigation
- Helps in threat detection
- Helps in compliance

STEP-BY-STEP: ENABLE CLOUDTRAIL

1

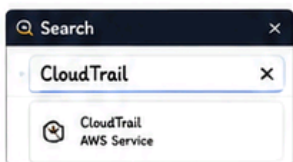
Go to AWS Management Console

aws Console

Open console and login to your AWS account.

2

Search for 'CloudTrail'



Type CloudTrail in the search bar and select it.

3

Click on 'Create trail'

Create trail

You will see the CloudTrail dashboard. Click on [Create trail](#).

4

Trail settings

- Trail name: [my-trail](#)
- Apply trail to all regions
- Create new S3 bucket
- Enable log file validation

Give a name and keep the default settings (better for beginners).

5

Review and Create

Create trail

Review everything and click on [Create trail](#).

6

Success!



Your CloudTrail is now enabled. Yay! 🎉

CloudTrail will automatically start recording API activities in your AWS account.



WHAT DOES CLOUDTRAIL RECORD?

- Who logged in (and from where)
- What actions were performed (like [creating a bucket](#))
- Changes made to resources (like [IAM](#), [EC2](#), [S3](#))
- Permission changes
- Deletion of resources

Everything is logged like a super hero memory! 🕵️



INDIAN STYLE JOKE 😄

You: Mom, why do we need CloudTrail?

Mom: Beta, without CloudTrail, we don't know who changed what!

You: So it's like ghar ka CCTV?

Mom: Exactly! Bas yahan chori data ki hoti hai! 😂



PRO TIP

Always keep CloudTrail ON in all environments (dev, test, prod). It is your [first line of visibility](#)!



COMMON MISTAKE



Don't forget to store CloudTrail logs in S3. Without S3, logs have nowhere to go! 😞



REMEMBER: Without logs, you are **BLIND**. With CloudTrail, you have **X-RAY VISION**!



Awesome! CloudTrail is now your Cloud CCTV. Let's generate some activity next!



STEP 3

CREATE IAM USER

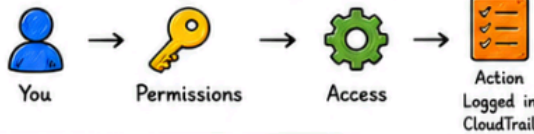
Let's create our own cloud student! 😎

Why IAM User?

In real companies, people don't use Root account daily. They use IAM Users (just like teachers 😊)

WHAT IS IAM?

IAM (Identity and Access Management) helps us create users and give them permissions.

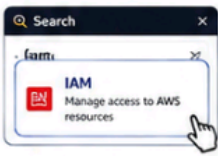


Indian Style Joke

You: Mom, I made new employee in cloud.
Mom: Salary kitna diya?
You: Zero! It's Free Tier IAM User! 😊

STEP-BY-STEP: CREATE IAM USER

1 Go to IAM



In AWS console, search for 'IAM' and open it.

2 Click Users



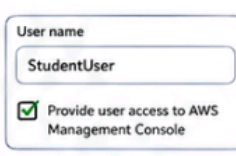
On left side menu, click on 'Users'.

3 Add User



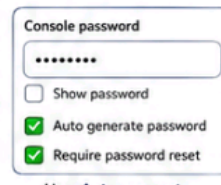
Click on 'Add users' button.

4 User details



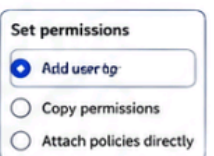
Enter User name: **StudentUser**
(You can give any name)
☒ Check the box below.

5 Set Password



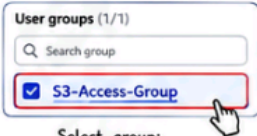
Use Auto generate password.
Keep both boxes checked. ☒ ☒

6 Permissions



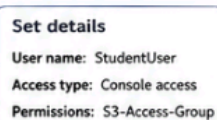
Choose: 'Add user to group'

7 Select Group



Select group: **S3-Access-Group**
(We will create this group now if not existing)

8 Review



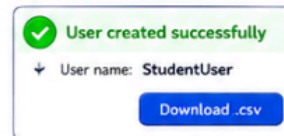
Check everything is correct.

9 Create User



Click on 'Create user'

10 Success!



User created! 😊
Download the credentials and save safely.

Note

- We will login using this user later.
- Never share this .csv file.
- Keep it safe like your exam hall admit card! 😊

STEP: CREATE GROUP (if not exists)

Go to IAM → User Groups → Create group

Group name: S3-Access-Group

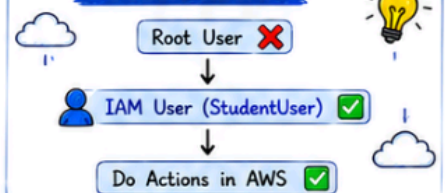
Attach policy: AmazonS3FullAccess

(Click Create group)

WHAT DID WE ACHIEVE?

- ☒ Created IAM User (StudentUser)
- ☒ Given permissions using a Group
- ☒ Ready to do actions in AWS (But not as Root user!)

QUICK RECAP



Next Step:

Let's login using this IAM user and generate some activity so that CloudTrail can record logs!

Remember: Don't use Root user for daily work. Always use IAM User!

STEP 4



GENERATE ACTIVITY

Let's do some actions in AWS so that CloudTrail can **record** them! 🎯



No activity
= No logs!
Let's create
some resources
and actions. 😊

WHAT KIND OF ACTIVITY?

We will perform different actions using our IAM User (**StudentUser**).



Create S3
Bucket



Upload
a File



Create
Another IAM
User



Change
Permissions



Login and
Logout

Think of it like this:

CloudTrail is the **CCTV**.
We are doing actions
in the building, so
CCTV can record. 😎

STEP-BY-STEP: GENERATE ACTIVITY

1 Login as IAM User

Account ID / Alias
123456789012

Username
StudentUser

Password
••••••••

Sign in

Login to AWS
Console using
StudentUser.

2 Create S3 Bucket



Give any bucket name
like **my-cloud-lab-123**
and create it.

3 Upload a File



Open the bucket and
upload a file.
(Any file is ok!)

4 Create Another IAM User



Create another user
name: **JuniorUser**
(Just for logs) 😊

5 Change Permissions

Attach policies

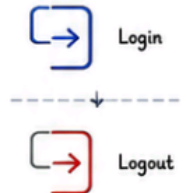
☐ ReadOnlyAccess

☒ S3FullAccess

☐ IAMReadOnlyAccess

Attach any policy
(like ReadOnlyAccess)
to JuniorUser.

6 Login - Logout



Login again and then
logout from AWS.
(This also creates logs)

WHAT HAPPENED NOW?

- ✅ You created a bucket
- ✅ You uploaded a file
- ✅ You created a new IAM user
- ✅ You changed permissions
- ✅ You logged in and logged out

All these actions are captured
by CloudTrail!



INDIAN STYLE JOKE

You: Bro, I created a bucket!
Friend: Bucket? For water? 😊
You: No yaar, S3 bucket!
Friend: Ohh... cloud wala balti! 😊
You: Haan! Usme file bhi daala!
Friend: Wah! Ab barsaat me
data overflow mat karna! 🌧️😭



Data sambhal ke
bhai! Overflow
na ho jaaye!

TIPS FOR YOU

- ✅ Use **only IAM Users** (not Root user).
- ✅ Do different actions so more logs
will be generated.
- ✅ Don't worry, if you make mistakes.
We are learning! 😊
- ✅ All these actions will help us in
Project 1 investigation. 🎯

FUN FACT

CloudTrail never sleeps! 😊
Even at 3 AM, it is recording
what you are doing! 🌙💻

Well Done!



You have generated activity in AWS.
Now CloudTrail has lots of logs to show us! 🎉



STEP 5

INVESTIGATE LOGS



Now let's open CloudTrail and see the **magic!** ✨



CloudTrail has all the recorded activities. Let's find clues! 🕵️

WHAT IS EVENT HISTORY?

Event History shows all the API calls and actions performed in your AWS account.

It shows:

- ✓ Event name (what action was performed)
- ✓ User (who performed it)
- ✓ Time (when it happened)
- ✓ Source IP (from where)
- ✓ And many other details!

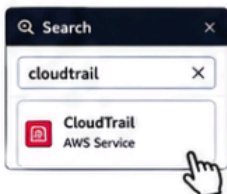


Think of CloudTrail like CCTV footage.

- 👤 Who came?
- ⚙️ What did they do?
- 🕒 When did it happen?
- 📍 From where?

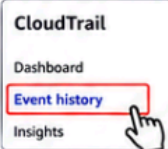
STEP-BY-STEP: VIEW AND INVESTIGATE LOGS

1 Open CloudTrail



In AWS Console, search for 'CloudTrail' and open it.

2 Go to Event history



On the left menu, click on 'Event history'.

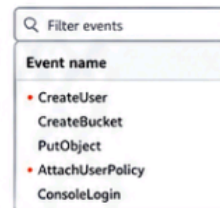
3 See the events

Event history (50+) Info
You can view the latest events.

Event history Download events

You will see a list of recent events recorded in your account.

4 Look for your actions



Find events like **CreateUser, CreateBucket, PutObject, ConsoleLogin** etc.

5 Check details

Event name: CreateBucket
Event time: May 11, 2024, 10:25:43 AM (UTC)
User name: StudentUser
Source IP: 103.25.10.45
Event source: s3.amazonaws.com
Request ID: 1a2b3c4d-5678...

Click on any event to see full details. This is your clue! 🕵️

6 Congratulations!



You just did your first cloud investigation! 🎉

SAMPLE EVENT HISTORY (EXAMPLE)

Event name	Event time (UTC)	User name	Source IP	Event source	Resources
ConsoleLogin	May 11, 2024, 09:15:21	StudentUser	103.25.10.45	signin.amazonaws.com	-
CreateUser	May 11, 2024, 09:20:11	StudentUser	103.25.10.45	iam.amazonaws.com	user/JuniorUser
CreateBucket	May 11, 2024, 09:25:43	StudentUser	103.25.10.45	s3.amazonaws.com	my-cloud-lab-123
PutObject	May 11, 2024, 09:30:10	StudentUser	103.25.10.45	s3.amazonaws.com	my-cloud-lab-123/hello.txt
AttachUserPolicy	May 11, 2024, 09:35:05	StudentUser	103.25.10.45	iam.amazonaws.com	policy/AmazonS3FullAccess

QUESTIONS TO ASK

- Who did this action?
- What action did they perform?
- When did it happen?
- From which IP address?
- Which resource was affected?



WHAT DID WE LEARN?

- ✓ CloudTrail records everything.
- ✓ Every action leaves a log.
- ✓ We can investigate who, what, when, where.
- ✓ Logs are our best friends in security!



INDIAN STYLE JOKE 😄

You: I checked the logs.
Friend: Kya mila?
You: Pata chala maine hi galti ki hai!
Friend: Toh log dekh kar sudhar jao.
You: Haan bhai, ab se smart work + log work!



Logs ne sab bata diya!



REMEMBER: Without logs, you are **BLIND**. With CloudTrail, you have **X-RAY VISION!** 👁️💡

PROJECT 1 COMPLETED



You just became a Cloud Security Detective! 🕵️

WHAT DID WE BUILD?



You created activity in AWS, CloudTrail recorded it, and you investigated it like a real detective!



WHAT DID WE LEARN?



CloudTrail
Cloud CCTV Camera

- Records everything happening in AWS.



IAM Users

Safer than Root User
Used by real companies

- Give least privilege access.
- Help teams work securely.



Logs

Cloud Memory

- Every action leaves evidence.
- Logs are your best friend.



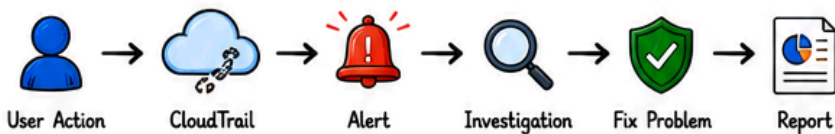
Investigation

Finding answers from logs

- Ask the right questions.
- Find the truth in seconds!



REAL LIFE CLOUD SECURITY ENGINEER WORKFLOW



This is how real SOC (Security Operations Center) analysts work!

WHAT QUESTIONS CAN WE ANSWER NOW?

- ✓ Who logged in?
- ✓ When did they login?
- ✓ Which IP Address?
- ✓ What action happened?
- ✓ Which AWS service was used?

CloudTrail gives us the answers! 💡

INDIAN STYLE JOKE 😄

Friend: Bro what are you doing?

You: Investigating logs.

Friend: Did someone hack NASA?

You: No bro.

I created the user myself. 😎😂



SKILLS UNLOCKED 🔓

- ✓ Cloud Monitoring
- ✓ IAM (Identity & Access Management)
- ✓ CloudTrail
- ✓ Log Analysis
- ✓ Investigation
- ✓ Critical Thinking



You are one step closer to becoming a Cloud Security Pro! 🚀

MISSION BADGE EARNED 🏆



Great job! Wear this badge with pride.



PROJECT 2 PUBLIC S3 BUCKET DATA LEAK LAB



WARNING

One wrong checkbox...
Can expose company data
to the entire internet! 🌐



Let's see how companies accidentally leak data... 👁️

PROJECT 2

PUBLIC S3 BUCKET DATA LEAK LAB



One wrong checkbox... can expose company data to the entire internet! 🐼



Many data leaks happen because of **wrong settings**.
Let's see how! 🤔

WHAT WE WILL DO IN THIS PROJECT?



WHY THIS MATTERS?

- ✓ S3 stores files like images, logs, backups, documents.
- ✓ If public, anyone on the **internet** can access!
- ✓ Real companies have leaked sensitive data this way! 🚨

WHAT IS AN S3 BUCKET?

S3 (Simple Storage Service) is AWS service to store data.

Think of it like a **cloud filing cabinet**.



HOW DATA LEAK HAPPENS?

- ✓ Create an S3 bucket
- ✓ Upload important files
- ✓ Change settings to "Public"
- ✓ Anyone can access the files

Accidentally done by many people! 🤔



WHAT CAN LEAK?

-  Customer data
-  Company documents
-  Source code
-  Backups
-  Logs & keys
- And more...



STEP-BY-STEP: PROJECT 2 OVERVIEW

- 1 Create S3 Bucket**

We will create a bucket to store our files.
- 2 Upload Files**

Upload a sample file (e.g., document or image).
- 3 Make Bucket Public (Mistake!)**

We will change the settings and make it public by mistake.
- 4 Become the Attacker**

From another browser, we will find and open the public file.
- 5 Fix the Problem**

We will block public access and follow best practices.
- 6 Verify & Secure**

Check again to make sure bucket is secure.

TOOLS WE WILL USE



WHAT WILL WE LEARN?

- ✓ How S3 works
- ✓ How public access causes data leaks
- ✓ How attackers find exposed buckets
- ✓ How to secure S3 the right way
- ✓ Real-world cloud misconfigurations

MISSION MODE: ON!



Let's break things (on purpose 😊) and then **fix** them like a real cloud security pro!

★ Remember: It's okay to break things in the lab, as long as we **learn** and **secure** them! 😎

Let's begin... 🚀

CREATE S3 BUCKET



First, we need a place to store data in the cloud! ☁️



WHAT IS AMAZON S3?

S3 (Simple Storage Service) is a storage service from AWS.

You can store:

- Images
- Documents
- Videos
- Backup files
- Logs
- ... and much more!



Your Cloud Bucket ☁️

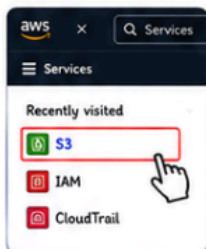
IMPORTANT TIP

Bucket name must be:

- ☒ Unique (only one in the world!)
- ☒ 3 to 63 characters long
- ☒ Lowercase letters, numbers, dots and hyphens only
- ☒ Cannot start or end with a dot or hyphen

STEP-BY-STEP: CREATE S3 BUCKET

1 Go to S3



Open AWS Console, search for "S3" and click on it.

2 Click Create bucket



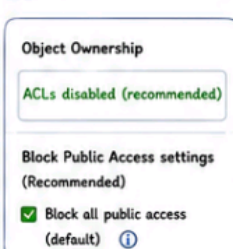
On the S3 dashboard, click the orange "Create bucket" button.

3 Bucket name



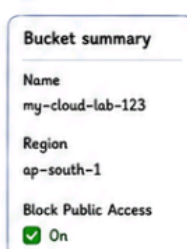
Enter a unique bucket name and select your AWS Region.

4 Leave default settings



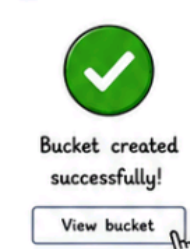
Keep the default settings for now. We will change later! 😊

5 Review



Review all details before creating your bucket.

6 Create bucket



Click "Create bucket". Congratulations! Your bucket is ready. 🎉

WHAT DID WE DO?

- ☒ Created an S3 bucket
- ☒ Gave it a unique name
- ☒ Chose a region (Mumbai)
- ☒ Kept it private for now



BUCKET NAME EXAMPLES

- ☒ my-cloud-lab-123
- ☒ company-backup-2024
- ☒ project2-data-store
- ☒ MyBucket (No capitals!)
- ☒ bucket@123 (No special chars!)
- ☒ .startsdot (No dots at start!)

WHERE IS YOUR BUCKET?

You can find your bucket in the S3 Dashboard.



QUICK ANALOGY



Physical Bucket → S3 Bucket

You put things in → Store safely
→ Access anytime, from anywhere! 🌐



Remember: Right now, your bucket is **PRIVATE**. Only you (your AWS account) can access it. In the next step, we will make a mistake and make it public 😬

FUN FACT

S3 is used by Netflix, Airbnb, Facebook, Disney+ and millions of other companies to store their data! 🎬

NEXT UP:

STEP 2 → MAKE BUCKET PUBLIC

(Oops... things will go wrong! 😬)





MAKE BUCKET PUBLIC

One checkbox... and anyone on the internet can see your data! 🐼



WHY THIS IS DANGEROUS?

- Your data becomes public
- Anyone can see, copy or download
- Hackers can find sensitive information
- Real companies have leaked data this way 😱

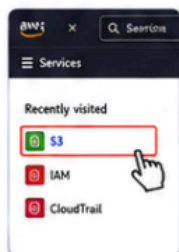


CAUTION!

- ✓ Do this **only** in labs
- ✓ Never do this in real life
- ✓ Delete the bucket after the lab
- ✓ We will fix it in the next step

STEP-BY-STEP: MAKE BUCKET PUBLIC

1 Open S3



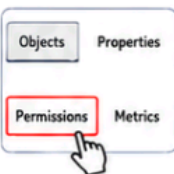
Open AWS Console, search for "S3" and click on it.

2 Select your bucket



Click on your bucket: **my-cloud-lab-123**

3 Go to Permissions



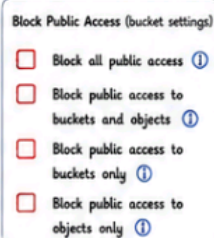
Click on the "Permissions" tab.

4 Edit Block Public Access settings



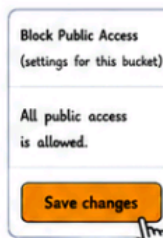
Scroll down to "Block Public Access" and click "Edit".

5 Uncheck all boxes



Uncheck ALL the boxes (one by one).

6 Save changes



Click "Save changes".

7 Confirm



Congratulations! Your bucket is now public 🎉

WHAT JUST HAPPENED?

- ✓ We allowed public access
- ✓ Anyone on the internet can list and read the files
- ✓ This is how data leaks happen!



WHAT CAN ANYONE DO NOW?

- See your bucket
- List your files
- Download your files
- Bad people can misuse it

REAL WORLD EXAMPLE

Many companies accidentally made their buckets public and leaked:

- Customer Data
- Source Code
- Backups
- Internal Documents



CHECK IT YOURSELF (TRY THIS!)

- Go to "Objects" tab.
- Upload a test file (e.g., **test.txt**).
- Open the file URL in a new tab.
- You should be able to see the file without any login! 🐼



HOW TO VERIFY IT'S PUBLIC?

You can use this format in browser:
<http://bucket-name.s3.amazonaws.com/>

Example:
<http://my-cloud-lab-123.s3.amazonaws.com/test.txt>
If the file opens, your bucket is public! 🐼

MISSION STATUS

- ✓ S3 Bucket Created
- ✓ Files Uploaded
- ✓ Bucket Made Public

MISSION 2.1 COMPLETE! 🚀



IMPORTANT REMINDER

You just learned how data leaks happen. Next step: We will become the attacker and see everything! 🐼

NEXT UP:

STEP 3 → BECOME THE ATTACKER

Let's see the world through an attacker's eyes! 👁️🐼

STAY CURIOUS, STAY SAFE! 🛡️

BECOME THE ATTACKER



Time to see how attackers find data leaks! 👁️👤



HOW ATTACKERS FIND LEAKS

- 🌐 Attackers scan the internet 24 x 7.
- 🔍 They look for open S3 buckets.
- 📄 If they find one, they list the files.
- ⬇️ Then, they download sensitive data.
- 👤 All this happens in minutes! 🐱



⚠️ WARNING!

- ✅ You just made your bucket public on the internet.
- ✅ Anyone can find it.
- ✅ Anyone can read your files.
- ✅ In real life, this can cause huge data leaks! 🚨

STEP-BY-STEP: BECOME THE ATTACKER

1 Open Incognito Window



Open a new Incognito (Window) in your browser.

2 Go to S3 Bucket Endpoint

<https://s3.amazonaws.com>

Enter this URL and press Enter.

<https://s3.amazonaws.com>

3 Search for your Bucket

my-cloud-lab-123 🔍

Use Ctrl + F (or Cmd + F) on Mac) and search for your bucket name.

Ctrl + F

4 Open the Bucket



If your bucket is public, you will see a link. Click on it!

5 List the Files



You can see all files inside your bucket **without any login!**

6 Download a File



Try downloading one file. You can do it because it's **public!**

WHAT WE JUST DID?

- ✅ We found a public S3 bucket on the internet.
- ✅ We listed the files.
- ✅ We downloaded the file.
- ✅ This is exactly what attackers do in real life! 🌐

WHAT CAN GO WRONG IN REAL LIFE?

- Customer data can be stolen
- Source code can be leaked
- Passwords/keys can be exposed
- Financial and legal damage
- Loss of trust and reputation



REAL WORLD EXAMPLE

Many companies have leaked millions of records because of one mistake: Making S3 bucket **public!** 🐼

CapitalOne AWS VERIZON & more...

"Attackers don't hack. They just find what we accidentally leave open!" 🎯

ATTACKER TOOLS (EXAMPLES)

- AWSBucketDump
- S3Scanner
- Bucket Finder
- Shodan
- Google Dorks



WHAT MAKES A BUCKET SAFE?

- ✅ Keep it Private
- ✅ Block Public Access ON
- ✅ Use IAM Roles & Policies
- ✅ Share only what is needed
- ✅ Always follow Least Privilege



TRY THIS (LAB FUN)

- ☐ Open your bucket in Incognito.
- ☐ List all files.
- ☐ Try downloading a file.

See how easy it is for attackers? That's why we must **fix** it! 🐼

REMEMBER

The internet never forgets! Once data is leaked, it spreads everywhere.



COMING UP NEXT

STEP 4 → FIX THE PROBLEM

We will lock the bucket and make it private again! 🔒

Be a hero. Fix the leak and save the data!



FIX THE PROBLEM



Let's lock the bucket and keep the data safe!



WHAT WE WILL DO?

We will remove public access and make our S3 bucket private again.



WHY THIS IS IMPORTANT?

- ✓ Prevents anyone from accessing your data.
- ✓ Protects customer information, company data and secrets.
- ✓ Follows security best practices.
- ✓ Keeps your cloud environment secure and trusted!

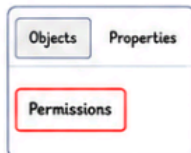
STEP-BY-STEP: FIX THE PROBLEM

1 Open S3



Go to S3 Console and select your bucket.

2 Go to Permissions



Click on the "Permissions" tab.

3 Block Public Access (Bucket Settings)

Block Public Access (settings for this bucket)

Edit

Click "Edit" under Block Public Access.

4 Enable All Options

- ✓ Block all public access
- ✓ Block public access to buckets and objects
- ✓ Block public access to buckets only
- ✓ Block public access to objects only

Check ALL the boxes and click "Save Changes".

5 Check Bucket Policy

Bucket policy

No policy
This bucket currently has no policy.

Make sure there is no public policy. (If yes, delete it.)

6 Verify It's Private



Your bucket is now private and secure!

WHAT DID WE CHANGE?

- ✓ We turned ON Block Public Access
- ✓ We removed public policies
- ✓ We stopped anonymous access
- ✓ Now only we (our AWS account) can access our data



BEFORE vs AFTER

BEFORE (BAD) AFTER (GOOD)

Public Bucket



Anyone on the internet can access

Private Bucket



Only you can access

HOW TO TEST?

- ☑ Open your bucket in Incognito mode.
- ☑ Try to open any file.
- ☑ You should see: "Access Denied"

ACCESS DENIED

REAL WORLD TIP

Always follow the principle of "Least Privilege"

Give access only to people who really need it, and only the access they need.



SECURITY CHECKLIST

- ✓ Block Public Access: ON
- ✓ Bucket Policy: No public access
- ✓ Objects: Private
- ✓ Only authorized users have access
- ✓ Data is safe and secure



YOU DID IT!

You fixed the problem like a real Cloud Security Engineer!

You just learned how to protect data in the cloud!



MISSION STATUS



Data Leak Fixed
Bucket Secured
You Rock!



MISSION 2.2 COMPLETE!

COMING UP NEXT

STEP 5 → VERIFY & SECURE

Let's double-check everything and make sure our bucket is 100% secure!

Security is not a one-time thing. Always verify, always protect!



PROJECT 2 COMPLETED

YOU JUST PREVENTED A REAL DATA LEAK!



Great job, Cloud Security Engineer! You made the cloud safe again! 



One wrong
checkbox could
have leaked
EVERYTHING!

WHAT WE DID IN PROJECT 2



SKILLS UNLOCKED

- ☒ S3 Basics
- ☒ Public vs Private Access
- ☒ Understanding Data Leaks
- ☒ Least Privilege Principle
- ☒ Real Cloud Security Thinking



STEP-BY-STEP RECAP

1 Create Bucket



We created an S3 bucket to store our files.

2 Made it Public



We changed the settings by mistake and made it public to the internet.

3 Attacker Found It



Anyone on the internet could find our bucket and access the files!

4 We Fixed It



We removed public access and made the bucket private again.

5 Verified & Secured



We checked the policy and confirmed the bucket is now secure.

6 Mission Complete



Mission 2 complete! Our data is safe and protected.

WHAT WE LEARNED

- ☒ S3 buckets are powerful but dangerous when exposed.
- ☒ Public access can cause serious damage in real life.
- ☒ Attackers don't need to hack — they just look for open buckets.
- ☒ Always follow least privilege and security best practices.



KEY PRINCIPLE

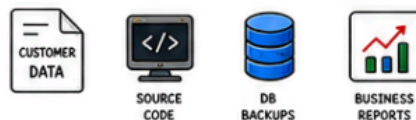
LEAST PRIVILEGE

Give access only to people who really need it, and only the access they actually need.



REAL WORLD IMPACT

Companies have lost millions of dollars because of public S3 buckets! 



Your small mistake in the cloud can become a big problem for the whole company!

CHECK YOURSELF

1. What happens when a bucket is public? 
2. How did the attacker find our bucket? 
3. What did we do to fix the problem? 
4. What is the best practice for S3 buckets? 
5. What principle did we follow? 

MISSION BADGE EARNED



CLOUD DATA PROTECTOR

You protected the data.
You secured the cloud.
You are awesome! 

CELEBRATE!



You turned a risky situation into a secure environment!
WELL DONE! 
You are thinking like a real Cloud Security Engineer!



WHAT'S
NEXT? 

PROJECT 3 → THREAT DETECTION LAB

Let's learn how to detect real threats in the cloud with GuardDuty! 

New Mission.
New Skills.
Let's go! 

PROJECT 3 – INTRODUCTION

INTRODUCTION TO THREAT DETECTION LAB



Now, let's learn how to detect real threats in the cloud!

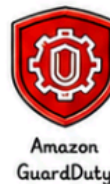


Time to become a Cloud Security Detective!
We will use Amazon GuardDuty!

WHAT IS AMAZON GUARDDUTY?

Amazon GuardDuty is an intelligent threat detection service that continuously monitors for malicious activity and unauthorized behavior in your AWS account.

- Uses Machine Learning and Threat Intelligence
- Finds threats you might miss
- Works across your AWS environment
- Always monitoring. Always protecting.



WHY DO WE NEED IT?

- ✓ Hackers are always scanning
- ✓ Threats can happen anytime
- ✓ Manual monitoring is difficult
- ✓ GuardDuty watches 24 x 7
- ✓ It alerts us when something suspicious happens

HOW DOES GUARDDUTY WORK?



Cloud Activity



GuardDuty Monitors



Detects Suspicious Activity



Generates Findings



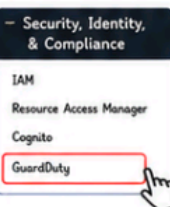
We Investigate & Take Action



Environment Stays Secure

STEP-BY-STEP: ENABLE GUARDDUTY (OVERVIEW)

1 Open GuardDuty



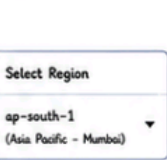
Search for "GuardDuty" in the AWS Console and click on it.

2 Enable GuardDuty



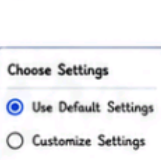
Click on the "Enable GuardDuty" button.

3 Choose Region



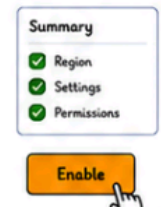
Select your Region (e.g., ap-south-1 Mumbai).

4 Choose Settings



For this lab, select "Use Default Settings".

5 Review & Enable



Review the summary and click "Enable".

6 Wait a Few Minutes



GuardDuty starts monitoring your environment.

7 See Your Findings



Go to "Findings" tab to see alerts when detected.

WHAT WILL WE DO?

- ✓ Enable GuardDuty
- ✓ Generate suspicious activity
- ✓ Investigate the findings
- ✓ Understand and respond



WHAT CAN IT DETECT?

- Unauthorized Access
- Malicious IP Connections
- Crypto Mining Activity
- Exposed Resources
- Compromised IAM Users and much more!



REAL WORLD USE

Companies use GuardDuty to detect and stop attacks early!



Detect early. Respond fast. Stay secure.

SKILLS YOU WILL BUILD

- ✓ Threat Detection
- ✓ Security Monitoring
- ✓ Incident Investigation
- ✓ Cloud Security Thinking
- ✓ SOC Analyst Skills



FUN FACT

GuardDuty analyzes billions of events per second using Machine Learning!



NEXT STEP

STEP 17 → ENABLE GUARDDUTY

Let's turn on our detective system!

Excited?
Let's detect our first threat!



ENABLE GUARD DUTY



Let's turn on our cloud detective! 🕵️



WHAT HAPPENS WHEN WE ENABLE GUARD DUTY?

- GuardDuty starts monitoring our AWS account.
- It looks for suspicious and malicious activity.
- If something looks bad, it creates a finding (alert).
- We get notified and can take action.
- Our environment stays secure 24 x 7!

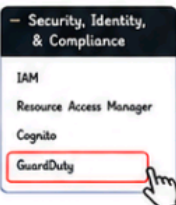


IMPORTANT!

- GuardDuty must be enabled in each region.
- It requires permissions to monitor your resources.
- It may take a few minutes to start showing findings.

STEP-BY-STEP: ENABLE GUARD DUTY

1 Open GuardDuty



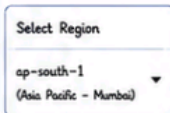
Search for "GuardDuty" in the AWS Console and click on it.

2 Click Enable GuardDuty



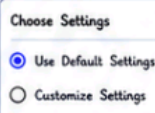
Click the "Enable GuardDuty" button.

3 Choose Region



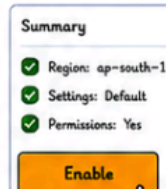
Select your Region (e.g., ap-south-1 Mumbai).

4 Choose Settings



For this lab, select "Use Default Settings".

5 Review & Enable



Review the summary and click "Enable".

6 Wait a Few Minutes



GuardDuty will now activate and start monitoring.

7 Confirmation



You will see a success message. Great job!

WHAT RESOURCES DOES IT MONITOR?

- EC2 Instances
- S3 Buckets
- IAM Users
- VPC Flow Logs
- DNS Logs
- and more...



It uses logs and machine learning to find threats!

EXAMPLE FINDINGS (ALERTS)

- Unauthorized API Calls
- Brute Force Login Attempts
- Malicious IP Connections
- Crypto Mining Activity
- Data Exfiltration Attempts



HOW IS IT DIFFERENT?

Traditional Monitoring	Amazon GuardDuty
Manual Rules	ML + AI Powered
Limited Visibility	Full AWS Environment
Detects known threats	Detects unknown threats
Human Dependent	24 x 7 Automated

GuardDuty is like having an AI security guard who never sleeps! 🕵️

BEST PRACTICES

- Enable in all regions
- Keep finding types ON (default is good)
- Review findings regularly
- Integrate with CloudWatch for alerts
- Respond quickly!



CHECK YOURSELF

- What is GuardDuty? ✓
- Why do we enable it? ✓
- What does it monitor? ✓
- What does it create? ✓

DID YOU KNOW?

GuardDuty uses Machine Learning and Threat Intelligence from thousands of AWS accounts worldwide!



MISSION STATUS

- GuardDuty Enabled
- Region Selected
- Monitoring Started

STEP 1 COMPLETE! 🚀



Our cloud detective is now ON DUTY! 🕵️

WHAT'S NEXT?

NEXT STEP → STEP 18 GENERATE THREAT ACTIVITY

Let's create some suspicious activity and see GuardDuty in action! 👁️

Exciting things are coming up! ⚡

GENERATE THREAT ACTIVITY

Let's create some suspicious activity and see GuardDuty detect it! 🤖👤



TYPES OF THREAT ACTIVITY WE CAN SIMULATE

- Brute Force Login Attempts**
Trying many passwords to login.
- Access Unusual Locations**
Login from an unexpected country.
- Use Unusual IAM Access**
Create new users or keys and use them.
- Exfiltrate Data**
Access or download large amounts of data.



GuardDuty
Detects & Alerts!

NOTE

- ✓ We are simulating for learning purposes only.
- ✓ Don't do this in real environments without permission.
- ✓ Always clean up after the lab.
- ✓ GuardDuty will help us find these threats quickly!

STEP-BY-STEP: GENERATE THREAT ACTIVITY

1 Open AWS Console



Go to <https://aws.amazon.com> and sign in to your AWS account.

2 Open IAM Console



Search for "IAM" in the console search bar and open it.

3 Create a Test User



Create a new IAM user (lab-user) with programmatic access.

4 Download Credentials



Download the Access Key ID and Secret Access Key. Save it securely! 🔒

5 Generate Suspicious Activity



Use the credentials in CLI/SDK to perform suspicious actions (e.g., many failed logins).

6 Wait & Check Findings



Wait a few minutes. GuardDuty will detect and show findings! 🕒

EXAMPLE: BRUTE FORCE LOGIN (USING CLI)

1 Open your terminal.



2 Run the following command multiple times with wrong passwords:

```
aws sts get-caller-identity \
--access-key <ACCESS_KEY> \
--secret-key <SECRET_KEY> \
```

💡 Tip: Run it 10-20 times quickly.

3 Wait 5-10 minutes.



4 Go to GuardDuty Findings and see the alert!



WHAT GUARD DUTY WILL DETECT

- Multiple failed logins
- Unusual API calls
- Logins from unusual locations
- New or unusual IAM access
- Potential data exfiltration

I see you! 🕒
Suspicious activity detected!



GuardDuty Alert

Finding: UnauthorizedAccess:IAMUser/...
Severity: **HIGH**
Type: Unauthorized Access
Resource: IAM User
Region: ap-south-1
First Seen: 2 minutes ago

BEST PRACTICES

- ✓ Always test in a lab environment.
- ✓ Enable MFA on accounts.
- ✓ Use least privilege access.
- ✓ Rotate keys regularly.
- ✓ Monitor findings daily.



CLEAN UP AFTER LAB

- ❑ Delete the test IAM user.
- ❑ Remove access keys.
- ❑ Revoke any temporary access.
- ❑ Keep your environment clean and secure!



PRO TIP

The more creative (but safe) your simulations are, the more you will learn about real-world threats and how GuardDuty protects you! 🤖

FUN FACT

GuardDuty uses Machine Learning, Threat Intelligence, and Behavior Analysis to protect millions of AWS accounts every day! 🌍



WHAT'S NEXT?

NEXT STEP → STEP 19 REVIEW FINDINGS

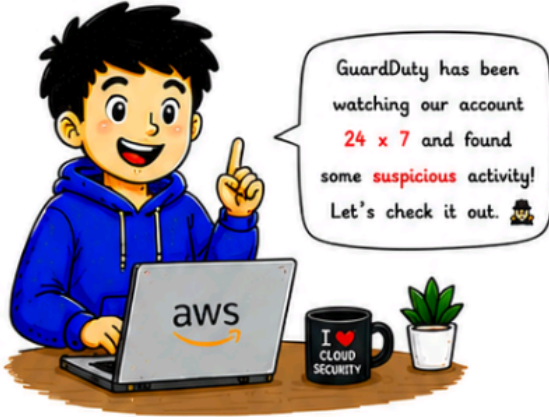
Let's open GuardDuty and review the findings it detected! 🕒

Nice work!
You just acted like a hacker (ethically) so we can stop real hackers! 🤖



REVIEW FINDINGS

Let's open GuardDuty and review the findings it detected! 👁️👤



WHAT IS A FINDING?

A finding is a security alert generated by GuardDuty when it detects something suspicious.

- Each finding has a severity.
- It contains details about the threat.
- It helps us decide what action to take. GuardDuty
- We can investigate, respond, and fix it.

WHY REVIEW FINDINGS?

- Understand what is happening in your account.
- Prioritize high severity threats.
- Take action to stop attacks.
- Improve your security posture.
- Stay safe and protect data!

STEP-BY-STEP: REVIEW FINDINGS

1 Open GuardDuty



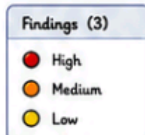
Go to the AWS Console and open GuardDuty.

2 Go to Findings



Click on "Findings" in the left menu.

3 View Findings List



You will see a list of findings with severity (High, Medium, Low).

4 Open a Finding

Finding ID:
1234abcd-5678-efgh
Severity: **HIGH**

Click on any finding ID to see more details.

5 Review Details

Type: UnauthorizedAccess:IAMUser/...
Resource: IAM User
Region: ap-south-1
First Seen: 2 min ago

Check the details like type, resource, region, and time. Understand the threat.

6 Investigate & Note



Take notes and decide what action to take next.

UNDERSTANDING SEVERITY

- HIGH** – Immediate attention needed. Could be a real attack.
- MEDIUM** – Should be investigated soon. Might be risky.
- LOW** – Informational. Monitor it.
- INFO** – General information only.



DETAILS YOU WILL SEE

- Finding ID
- Finding Type (what happened)
- Resource (what was affected)
- Region
- First & Last Seen
- Severity
- Description (what GuardDuty found)

EXAMPLE FINDING

Finding ID: 6f2a7b9c-1234-4abc-9def-56789abcde01
Type: UnauthorizedAccess:IAMUser/ConsoleLogin
Resource: IAM User – lab-user
Region: ap-south-1
First Seen: 3 minutes ago
Severity: **HIGH**
Description: GuardDuty detected a console login using incorrect password from an unusual location.



WHAT TO LOOK FOR

- Unusual login locations
- Brute force login attempts
- Unauthorized IAM access
- Malicious IP addresses
- Data exfiltration

WHAT TO DO NEXT

- Investigate the details
- Verify if it's a real threat
- If real, take action immediately
- Update security settings
- Monitor for more activity

BEST PRACTICES

- Review findings regularly
- Act on high severity first
- Use least privilege access
- Keep your environment secure
- Enable all GuardDuty data sources

DID YOU KNOW?

GuardDuty uses Machine Learning and Threat Intelligence from billions of events to find threats you might miss!



WHAT'S NEXT?

NEXT STEP → STEP 20 INVESTIGATE A FINDING

Now that we found a threat, let's investigate it like a pro detective! 👤🔍

You are doing awesome!
Let's dig deeper!



WRITE INCIDENT REPORT

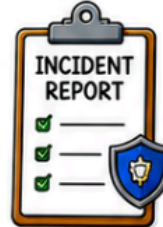


Good detectives document everything! 🕵️👉



WHY WRITE REPORTS?

- ✓ Helps track what happened
- ✓ Supports incident response and fixes
- ✓ Provides evidence for audits
- ✓ Improves security and processes
- ✓ Shares knowledge with the team



REMEMBER!

Security teams document everything. 🕵️

Good reports =
Stronger security! 🛡️

CREATE A SIMPLE REPORT

1 What happened?



Describe the incident in simple words.
What did you see?

2 When?



Date and time it happened.

3 Impact?



Who or what was affected?
What was the impact?

4 Fix?



What did you do to fix it?
What is the status now?

5 Anything else?



Add any extra details, screenshots, logs, or notes.

A FUNNY REMINDER 😄

No report?



No proof.



No proof?



Manager asks questions.



So...



Write the report!

WHAT TO INCLUDE

- ✓ Clear title
- ✓ Incident ID (if available)
- ✓ Detailed timeline
- ✓ Actions taken
- ✓ Outcome / status
- ✓ Attachments (logs, screenshots)



EXAMPLE: REPORT SNAPSHOT

- ✓ Title: Multiple Failed Login Attempts Detected
- ✓ Date/Time: 2024-05-10 14:35 UTC
- ✓ Impact: Possible unauthorized access attempt to IAM user.
- ✓ Actions Taken: Investigated, reset password, enabled MFA.
- ✓ Status: Resolved
- ✓ Notes: No data was accessed.
- ✓ Attachments: log-sample.txt, screenshot.png



BEST PRACTICES

- ✓ Be clear and factual.
- ✓ Don't guess—use data!
- ✓ Write as if others will read it.
- ✓ Keep it short and useful.
- ✓ Save and share the report.



WHAT'S NEXT?

NEXT STEP → STEP 5 RESPOND & REMEDIATE

Now that we reported it, let's take action and fix the problem! 🛠️🛡️

Reports may not stop attacks, but they help us stop the next one! 😊



THREAT DETECTED. INCIDENT INVESTIGATED. MISSION COMPLETE. ✓



What a journey! We did it! Great work! 😊

WHAT WE DID



WHAT DID WE LEARN?



THREAT DETECTION

Finding suspicious activity automatically.



SECURITY MONITORING

Watching cloud resources continuously.



INCIDENT INVESTIGATION

Understanding what happened.



INCIDENT REPORTING

Documenting everything.

SKILLS UNLOCKED

- ✓ GuardDuty
- ✓ Threat Detection
- ✓ Cloud Monitoring
- ✓ Investigation
- ✓ Reporting
- ✓ SOC Analyst Thinking



REAL SECURITY WORKFLOW



A SMALL JOKE 😄



ALERT APPEARS...



BEGINNER:

Something happened.



ANALYST:

What happened?



SENIOR ANALYST:

Let's check the logs. 😂

MISSION BADGE



REMEMBER

- ★ Detect early.
- ★ Investigate smart.
- ★ Respond fast.
- ★ Document always.
- ★ Stay secure & keep learning! 🔒

Congratulations. 🎉

You completed all

3 PROJECTS!



Keep building. Keep learning. Keep securing! 🚀



You're now one step closer to becoming an **awesome** Cloud Security Engineer! 😊



NEXT ADVENTURE: Keep practicing, keep exploring, keep securing the cloud! ☁️

YOU STARTED AS A BEGINNER. NOW YOU THINK LIKE A CLOUD SECURITY ANALYST.



What a journey! Let's look at what we accomplished! 🧐

PROJECT RECAP

PROJECT 1

CLOUD MONITORING LAB



Learned:

- ✓ CloudTrail
- ✓ Logs
- ✓ Investigation
- ✓ Cloud Activity Monitoring

We monitored cloud activity and investigated events.

PROJECT 2

DATA LEAK LAB



Learned:

- ✓ S3 Basics
- ✓ Misconfigurations
- ✓ Public Access Risks
- ✓ Data Protection

We found and fixed data leaks before attackers could use them.

PROJECT 3

THREAT DETECTION LAB

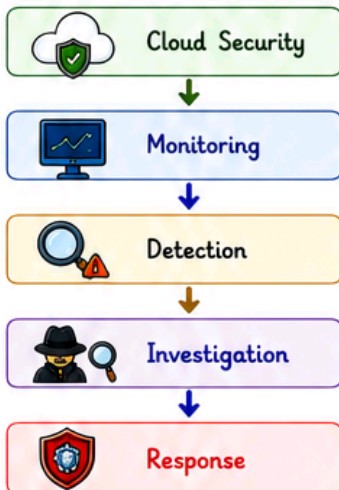


Learned:

- ✓ GuardDuty
- ✓ Threat Detection
- ✓ Alert Investigation
- ✓ Incident Reporting

We detected threats, investigated them, and wrote incident reports.

SKILLS LEARNED



This is how security professionals think and work every day! ★

WHAT CLOUD SECURITY ENGINEERS DO DAILY

- Monitor Alerts
Watch for suspicious activity.
- Investigate Incidents
Find out what happened.
- Protect Data
Keep data safe and secure.
- Fix Misconfigurations
Reduce risks and close gaps.
- Write Reports
Document findings and actions.

★ Security is not a tool.
It's a mindset. 🧠

CAREER PATHS

- SOC Analyst**
Monitor, detect, and respond to threats.
- Cloud Security Analyst**
Secure cloud environments and investigate issues.
- Security Engineer**
Build security controls and automation.
- Incident Responder**
Respond quickly to security incidents.
- Blue Team Analyst**
Defend systems and protect organizations.

Many paths. One mission:
Keep systems safe! 💪



Remember:

You don't need to be a genius.
You just need curiosity,
consistency, and passion!
Keep going! 🚀

PRO TIP

Keep practicing in your lab.
Try new things. Break things.
Fix things. That's how you
become a pro! 💻 ⚙️



YOU DID IT!

You completed all
3 projects and learned
real-world cloud
security skills! 🏆

“ Learning by doing beats learning by watching. ”

– Keep building. Keep learning. Keep securing. 🔒



YOUR CLOUD SECURITY LEARNING ROADMAP

Keep learning. Keep building. Keep leveling up! 🤖🚀

Here's the
roadmap to become
an **amazing Cloud
Security Engineer!** 🔥



THE 10 LEVEL ROADMAP



LEVEL UP TIPS

- ✓ Practice in your lab
- ✓ Break things (safely)
- ✓ Read documentation
- ✓ Build projects
- ✓ Join communities
- ✓ Ask questions
- ✓ Never stop learning!



REMEMBER

Every expert was
once a beginner.
Consistency beats
talent! 💪



YOUR GOAL



Protect systems.
Defend users.
Secure the cloud.
Make an impact!



FUN FACT

Cybersecurity is like a
game. The more you learn,
the more powerful you
become!



One step at a time.
One skill at a time.
You will get there!



NEXT MISSION BEGINS NOW!



The cloud is yours. Go build. Go secure. Go make a difference!



START YOUR CYBERSECURITY JOURNEY TODAY

The best way to predict the future is to build it. 😊

You've built the foundation.
Now build your future. ★



YOU COMPLETED:



CLOUD MONITORING

You monitored activity like a pro. ✓



DATA LEAK PREVENTION

You secured data and closed the leaks. ✓



THREAT DETECTION

You detected threats and investigated them. ✓

NOW CONTINUE BUILDING:

- ✓ Cloud Security Skills
- ✓ SOC Analyst Skills
- ✓ Detection Skills
- ✓ Incident Response Skills
- ✓ Investigation Skills
- ✓ Reporting Skills
- ✓ Problem Solving Mindset
- ✓ Real-World Experience



BLUETEAMERS.IO

Learn Cybersecurity Through
Hands-On Practice



PRACTICE
Real Labs



LEARN
Step-by-Step



BUILD
Real Skills



GROW
Your Career

WHY HANDS-ON MATTERS



Practice builds confidence.



Labs build understanding.



Experience builds expertise.



You become job-ready!

Keep practicing. Keep improving.
Keep winning! 🏆

JOIN THE COMMUNITY



Learn with thousands of blue teamers.



Share knowledge.



Ask questions.



Grow together.

You're not alone on this journey! 🤝

WHAT'S NEXT?

- ➔ Build more labs
- ➔ Solve more challenges
- ➔ Gain real experience
- ➔ Get certified
- ➔ Land your dream job
- ➔ Make an impact in cybersecurity!



REMEMBER:

Every expert was once a beginner.
Every journey starts with the first step.
You've already taken it. Keep going! ★

PERSISTENCE FORMULA



Practice



Consistency



Curiosity



Success



Keep going!
Your future self will thank you! 😊

“Weekend projects create weekday opportunities.”

BUILD. DEFEND. IMPACT.

SEE YOU IN THE SOC! 😊

